TRGBCurves (Bundle 3.2)

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General Description

TRGBCurves is a visual component that allows the creation, displaying and manipulation of tonal curves for image processing. By defining a curve for each Color channel and the Luminance channel of a Full Color picture, you will be able to define a correction for each tone of the channel range (0..255), therefore applying different amounts of change for each individual tone. Curves correction is a feature found in professional image and photo editors like Adobe Photoshop.

The component can apply the curves correction to both TBitmap and TIEBitmap encapsulated pictures. *TIEBitmap is the class used by the ImageEn graphics library. You can purchase a license of ImageEn at www.imageen.com

Features

If you are using the ImageEn library you can preview the changes and apply the curves directly on a TImageenView Component.

Very little coding is required from your side and the entire process of previewing and applying changes is handled automatically.

1· Edits each color channel and the luminance channel separately. Supports 8bits grayscale and full-color 24/32 bits TIEbitmap and TBitmap.

2· The user can add / remove / move the points that define the curve, using the mouse. The program can also do this by code.

3· A OnChangeCurve event notifies whenever something about the curves has changed.

4· Curves can be saved to a text file or to a Ini file and reloaded. From v. 3.2 support for loading and saving to Photoshop curves file format was added.

5· Curves may also be defined by mathematical formulas

6· In addition to displaying curves and applying them, the component can get the histogram from any image and graph it together with the curves.

7· When you register TRGBCurves to work together with a TImageenview component, any change to the curve will be previewed on the imageEn viewer component. Then you can apply the changes by using the provided methods of TRGBCurves.

8· It Integrates perfectly with ImageEn Layers and Selections as well.

Demo Project

The demo project included contains all the most important features of TRGBCurves. Please refer to the code to quickly learn how to use the component.

Included Files

NWSComps_RGBCurves.pas
Contains the main class derived from TCustomControl

NWSComps_RGBCurves_Types.pas
Contains Types definitions

NWSComps_RGBCurves_Math.pas
Defines the ,mathematics needed to build the curves

NWSComps_IEUtils_Previews.pas
Defines the ,classes needed to perform real time preview on a TImageEnView component

NWSComps_RGBCurves.inc
Contains Compiler Directives Definitions

**********************NWSComps_RGBCurves_Types**********************

TRGBCurves_Layout = class(TPersistent) //This class is not described here because it is used by the TRGBCurves Component in its properties and described later

TRGBCurves_CanMoveType = (cm_XY, cm_X_only, cm_Y_only, cm_CannotMove);

TRGBCurves_Channel_ViewMode = (fvmAll, fvmRed, fvmGreen, fvmBlue, fvmRGB);

TRGBCurves_FormulaCurve = function(x: single): single of object;

TRGBCurves_Editmode = (emNil, emMovePoint, emAddPoint);

TRGBCurves_Histogram = record active: boolean;

DisplayMode: TRGBCurves_HistogramDisplayMode; ScaleMode: TRGBCurves_HistogramScaleMode; RGBMode: TRGBCurves_HistogramRGBMode; Reds: array[0..255] of cardinal;

**Greens: array[0..255] of cardinal; Blues: array[0..255] of cardinal;
Grays: array[0..255] of cardinal;
peakred, peakgreen, peakblue, peakgray: cardinal; avgred, avggreen, avgblue, avggray: cardinal;**

TRGBCurves_HistogramDisplayMode = (HDmFilled, HDmTransparent, HDmLines);

**TRGBCurves_HistogramRGBMode = (HmRed, HmGreen, HmBlue, HmGray, HmAll);
TRGBCurves_HistogramScaleMode = (HSmLinear, HSmLog, HSmauto);**

TRGBCurves_LUT = array[0..255] of byte;

TRGBCurves_Lutarray = array[0..3] of TRGBCurves_LUT;

TRGBCurves_Mode = (cmFormulaCurves, cmBuildCurves, cmNoCurves); //cmNoCurves is used for viewing just the histogram

TRGBCurves_MovePointEvent = procedure(sender: TObject;

CurrentPoint: TPoint; CurrentPoint_Idx: integer;
var CanMove: TRGBCurves_CanMoveType) of object;

TRGBCurves_Options =

Set of (Opt_Allow_AddPoints,
Opt_Allow_RemovePoints,
Opt_Allow_VinculatedPoints);

TRGBCurves_RGBmode = (cmAll, cmRed, cmGreen, cmBlue, cmRGB);

TRGBCurves_RGBSpace = (csRGB, csLum, csMixed);

TRGBCurves_Row = array[0..30000] of byte; tpRGBCurves_Row = ^TRGBCurves_row;

**********************NWSComps_RGBCurves**********************

Class TRGBCurves is a descendant of TCustomControl

Properties

CurveMode: TRGBCurves_Mode

This property can have 2 values:

1· **cmFormulaCurves**
2· **cmBuildCurves**

If you set it to cmFormulaCurves the component will show mathematical functions that you assign using the FormulaCurve_Red, …, FormulaCurve_Lum properties.

If you set it to cmBuildCurves the component will use interpolation to fit the points you provide by the interface in a similar way to the Photoshop curves.

Complementary Curves

When this option is enabled, the curves will be converted into their complementary curves, that is: the Red curve will become the Cyan curve, the Green curve will become the Magenta curve, the Blue curve will be the Yellow curve, and the Luminance curve will become the Black (or Ink) curve.

RGBMode: TRGBCurves_RGBMode
This property can have 4 values:

1· **cmAll**
2· **cmRed**
3· **cmGreen**
4· **cmBlue**

If you set RGBMode to any channel, you will see only the selected channel function/curve.
If the mode is cmAll you will see all the functions if you are in cmFormulaCurves curve mode or you will see the luminance channel if you are in cmBuildCurves curve mode.

RGBSpace: TRGBCurves_RGBSpace

This property can have 3 values:

1· csRGB
2· csLum
3· csMixed

1- In RGB space, when changing the color channels, the Luminance will not be taken into account.

2- In Lum space the Luminance is taken into account, when changing the color channels. This space may tend to oversaturate the colors when brightness is increased, and to overdesaturate the colors when brightness is decreased.

3- In Mixed space a combination of the two spaces above is used. The mixed amount is established by the ColorSpaceMixAmount property

ColorSpaceMixAmount: integer
0-100
Determines the amount of mixing between RGB Space and Lum Space

ExportDirectory:string

Default Folder's Name that shall be displayed in the Save Dialog when the user exports the curves to a file.

UseGDIPlus: Boolean
If true the component will try to use GDIPlus to draw the graph elements. Requires ImageEn to work.

UseFastPreview: Boolean
It enables / disables fast preview when working with TImageEnView.
Use this if you are dealing with big pictures. It will calculate a less-accurate preview of the current changes, speeding up the preview process.

FastPreviewMode: TIEUtils_IEPreview_Mode
Can be set to

pm_FullUpdate: use this when you have layers
pm_Flat: use this only if you do not have layers
pm_Auto: let the preview engine decide which mode to use. Best Choice usually.

HistogramDisplayMode: TRGBCurves_HistogramDisplayMode

This property can have 3 values:

HdmFilled
HdmTransparent
HDmLines

It specifies the style of the Histogram

HistogramRGBMode: TRGBCurves_HistogramRGBMode
This property can have 5 values:

1· HmRed

2· HmGreen

3· HmBlue
4· HmGray
5· HmAll

The first 3 values represent the histogram for each color channel, the HmGray value represents the luminance channel and the HmAll will show all channels alltogether.

ShowHistogram
Boolean
Default Value is FALSE.

To show an histogram, you need to create one from a Tbitmap or a TIEBitmap using the GetHistogramfromBMP orGetHistogramfromIEBMP methods

GraphBackColor: TColor

You can choose the color of the Graph Background. Default is White.

BorderPercent: single (0-100)
It's the size of the border around the Graph Area, in percent. IT changes the BorderFixed property as well.

You can adjust it to enable more room for the graph labels.

BorderFixed: Cardinal

It's the size of the border around the Graph Area.

You can adjust it to enable more room for the graph labels.

CurveColor_Lum: TColor

You can choose the color of the Luminance curve. Default is Black.

CurveColor_Red: TColor

You can choose the color of the Red curve. Default is Pure Red.

CurveColor_Green: TColor

You can choose the color of the Green curve. Default is Pure Green.

CurveColor_Blue: TColor

You can choose the color of the Blue curve. Default is Pure Blue.

GridColor: TColor

You can choose the color of the Grid Lines.

GridMedianLine_Color: TColor

You can choose the color of the median line of the grid.

LineSize: Cardinal

It's the size of the Line of the curve

LineOpacity: byte
0-255
Only works when GDIPlus is used (UseGDIPlus = TRUE).

You can set the amount of color opacity of the Line of a curve

PointSize: Cardinal

It's the size of the control Point of the curve when Curve Mode =cmBuildCurves

PointOpacity: byte
0-255
Only works when Curve Mode =cmBuildCurves and GDIPlus is used (UseGDIPlus = TRUE).
You can set the amount of color opacity of the inner part of a control point of a curve

PointFillStyle: TBrushstyle
Only works when Curve Mode =cmBuildCurves .

You can set the style of the inner filling of a control point of a curve

ShowVerticalCaptions: Boolean
You can hide/show the labels for the Y axis of the graph.
This is useful if you want to display an histogram without the curves, since the histogram has no vertical labels.

ShowMedianline: Boolean
Default is TRUE.

The median dash-dotted line is displayed to visualize the discard of the curve from the no-change position.

Font: Tfont

Choose the font to display text in the graph.

Points: TRGBCurves_Pointslist

The list of Currently Displayed points

InterpLinear_Min: integer
Default is 15.

Changes the least amount of local linearization that should be applied to the interpolated curve. Local linearization makes the curve shape more stable, however it tends to segment the curve.

InterpLinear_Max: integer
Default is 65.

Changes the maximum amount of local linearization that can be applied to the interpolated curve.Local linearization makes the curve shape more stable, however it tends to segment the curve.

MinFittingX: Integer (Read only)

First X Coordinate for which the Curve value is Minimum (Y=0).

MaxFittingX: Integer (Read only)

First X Coordinate for which the Curve value is Maximum (Y=255).

FormulaCurve_Red: TRGBCurves_FormulaCurve

Assign to this property the function that you want to use for the RED channel when Curve Mode = cmFormulaCurves

FormulaCurve_Green: TRGBCurves_FormulaCurve

Assign to this property the function that you want to use for the GREEN channel when Curve Mode = cmFormulaCurves

FormulaCurve_Blue: TRGBCurves_FormulaCurve

Assign to this property the function that you want to use for the BLUE channel when Curve Mode = cmFormulaCurves

FormulaCurve_Lum: TRGBCurves_FormulaCurve

Assign to this property the function that you want to use for the LUMINANCE channel when Curve Mode = cmFormulaCurves

Methods

ApplyCurvesToBitmap
Parameters:

thebitmap : Tbitmap

Apply the Curves to the Tbitmap **(Pixel format must be supported.)**

ApplyCurvesToIEBitmap
Parameters:

theIEBitmap : TIEbitmap
Mask:TIEMask
EditeRect:TRect
bUseMask: boolean

Apply the Curves to the TIEbitmap **(Pixel format must be supported.)**

The Mask is a ImageEn Selection that defines any area inside the TIEBitmap in which the changes should be applied.

The EditedRect is the rectangular crop area in which the changes should be applied

bUseMask defines whether to consider the Mask or ignore it.

ApplyCurvesToImageenView

parameters:
theieview : TimageenView

Apply the Curves to a TImageEnView of choice. Current layer will be affected. If a selection (TIEMask) is defined only selected parts will be affected by changes.
Pixel format must be supported.

IEView_Preview_Register
Parameters:

theieview : TimageenView

Registering a TImageenView component to work in conjunction with TRGBCurves will automatically reflect any changes made to the curves on the registered TImageEnView display.

The preview will affect the current layer and will take care of any selection (if present). Every changes in the display (zooming, panning, selecting layers, creating selections) will instantaneously update the preview.

IEView_Preview_UnRegister
Parameters:
None

After a TimageenView component has been registered for preview by the IEView_Preview_Register method, you will need to unregister it by this command. Unregistering the component will stop TRGBCurves from updating the TImageenview display.

IEView_Preview_ApplyChanges
Parameters:

theieview : TimageenView

Apply the Curves to the TImageEnView that was registered for the previews. The last changes previewed will be applied on the current layer and If a selection (TIEMask) is defined only selected parts will be affected by changes.

GetHistogramfromBMP
Parameters:

thebitmap : Tbitmap

Loads the histogram from a Tbitmap object.

GetHistogramfromIEBMP
Parameters:

theilEbmp : TIEBitmap

Loads the histogram from a TIEBitmap object.

GetHistogramfromImageEnView
Parameters:

theieview : TimageenView

Loads the histogram from a TimageenView object.

ExportCurves
Parameters:
None

Opens the save dialog to export curves to text files.

ImportCurves: Boolean
Parameters:
None

Opens the open dialog to import curves from text files. Returns True if any curve is imported.

SaveCurves_ToFile
Parameters:

theCurvesfile: string

Saves the curves to the specified file name.

LoadCurves_FromFile
parameters:

theCurvesfile: string

Load the curves from the specified file name

PS_SaveCurves_ToFile
Parameters:

theCurvesfile: string

Saves the curves to the specified file name in Photoshop format.

PS_LoadCurves_FromFile: Boolean
Parameters:

theCurvesfile: string

Load the curves from the specified file name in Photoshop format

ExportCurvesToINI
Parameters:

Inifile: Tinifile
IniSection: string

Exports curves to INI files.

ImportCurvesFromINI
Parameters:

Inifile: Tinifile
IniSection: string

Imports curves from INI files.

ResetPoints
Parameters:
none

Reset all curves by resetting the values of all the control points to default.
Only works with Curve Mode = cmBuildCurves

SetthePoint
Parameters:

ix: integer
thepoint: TRGBCurves_doublepoint

Set the value of the point at the position ix of the current curve Only works with Curve Mode = cmBuildCurves

SetCurrentPoint
Parameters:

thepoint: TRGBCurves_doublepoint

Set the value of the current point (the one that was last selected, that is moved or created) of the current curve
Only works with Curve Mode = cmBuildCurves

GetCurrentPoint: TRGBCurves_double
Parameters:
none

Get the current point (the one that was last selected, that is moved or created) of the current curve

Only works with Curve Mode = cmBuildCurves

GetCurrentPoint_Idx: integer
Parameters:
none

Get the Index of the current point (the one that was last selected, that is moved or created) of the current curve. Only works with Curve Mode = cmBuildCurves

SetCurrentPoint_Idx
Parameters:

Set the Index of the current point (the one that was last selected, that is moved or created) of the current curve. Only works with Curve Mode = cmBuildCurves

AddPoint
Parameters:

thepoint: TRGBCurves_doublepoint

Add a point to the curve. Only works with Curve Mode = cmBuildCurves

RemovePoint

Parameters:
ix: integer

Remove the point at the position ix of the current curve. Only works with Curve Mode = cmBuildCurves

RemovePoint_Current
Parameters:
None

Remove the current point (the one that was last moved or created) of the current curve. Only works with Curve Mode = cmBuildCurves

PointExists: boolean
Parameters:

const thepoint: TRGBCurves_doublepoint

Returns TRUE if the specified point already exists in the current curve. Only works with Curve Mode = cmBuildCurves

ExportLUT: TRGBCurves_LUT Func
Parameters:

mode: TRGBCurves_RGBMode

The result of this function is a Lookup Table. It contains the ordered curve values for each of the [0--255] tones of a channel. The channel is specified by the Mode

ExportLUTs
Parameters:

var BGRALUTarray: TRGBCurves_Lutarray
var GrayLUT: TRGBCurves_LUT

The returned parameters contain the Lookup Tables for all channels. GrayLut is the luminance channel.

Events

OnChangeCurve: TNotifyEvent
Use this event to be notified whenever a change happens to the curves.

OnChangeRGBMode
Use this event to be notified whenever the RGBMode changes

OnMovePoint: TRGBCurves_MovePointEvent
Use this event to be notified whenever a Point has been moved

OnMovingPoint: TRGBCurves_MovingPointEvent
Use this event to be notified whenever a Point is being moved

OnAddPoint: TRGBCurves_Point_Event
Use this event to be notified whenever a Point has been added

OnRemovePoint: TRGBCurves_Point_Event
Use this event to be notified whenever a Point has been removed

OnCustomDraw_Labels_HorzAxis: TRGBCurves_CustomDrawLabelEvent
Use this event to be notified whenever the Labels of the horizontal axis are redrawn

OnCustomDraw_Labels_VertAxis: TRGBCurves_CustomDrawLabelEvent
Use this event to be notified whenever the Labels of the vertical axis are redrawn

OnCustomDraw_Point: TRGBCurves_CustomDrawPointEvent
Use this event to be notified whenever a Point is redrawn

OnIEView_BeforePreview: TnotifyEvent
Use this event to be notified before the preview is displayed on the Imageenview component

OnIEView_AfterPreview: TnotifyEvent
Use this event to be notified after the preview is displayed on the Imageenview component